

Trigger instruction manual

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1 What is trigger

The camera supports two output modes, namely continuous and trigger mode. Trigger is a passive mode. In the trigger mode, the camera enters the ready state. When the trigger signal is generated, the camera immediately starts exposure and then outputs the image. Triggering is generally used in occasions that require high real-time performance, such as capturing high-speed moving objects.

2 How to generate a trigger signal

In the trigger mode, the trigger signal is the only condition for starting the camera exposure. The trigger signal can be generated by software or by external hardware (mechanical switch/electronic sensor/PLC/microcontroller/IO card). We call these two methods software trigger and hardware trigger.

2.1 Software trigger

In the SDK, there is a dedicated interface for generating software trigger signals, and you can also set the number of frames for a trigger. The following figure is an example of triggering in Demo

Exposure	Color Process	NTSC (Auto Up Table)	Graphic
Video	Resolution	Trigger	Crosshairs
Device			

Mode

☐ Continuous

☒ Software Trigger once

☐ Hardware

Parameters of software trigger

Frames at one Apply

Parameters of external trigger

Mode: Falling edge Shutter mode: Standard

Jitter apply

Delay(us) apply

Strobe signal

☒ Auto

☐ Semi-automatic

Effective ☒ LOW ☐ HIGH

Delay(us) Apply

Pulse Apply

Camera configuration parameters archive

Groups: ☒ A ☐ B ☐ C ☐ D Save Default

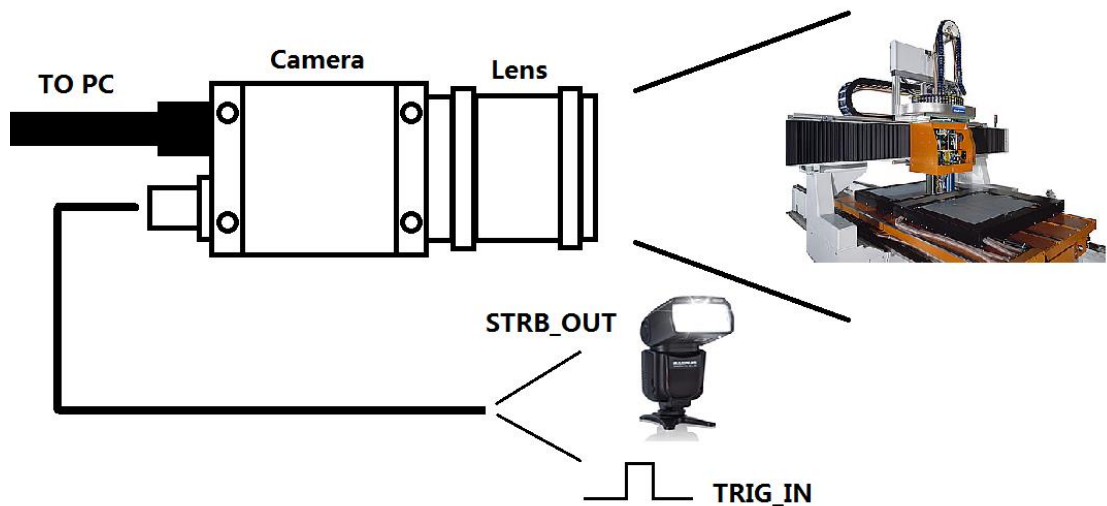
Save parameters to the specified file Load from the specified file

Load mode: ☒ ByModel ☐ ByName ☐ BySN

☒ Save on switch group ☒ Save on exit program

2.2 Hardware trigger

There is an aviation head connector on the camera, which is used to connect an external switch and flash. A typical external trigger application is shown in the figure below:



The external trigger input terminals are labeled TRIG+ and TRIG- (please find the line sequence definition table for the corresponding line colors). The camera uses optocoupler isolation and can identify rising edge, falling edge, double edge, high level, and low level. The methods for generating these signals are as follows:

A rising edge: when the voltage across TRIG+ and TRIG- changes from 0V to 5~24V, it changes once;

A falling edge: when the voltage across TRIG+ and TRIG- changes from 5~24V to 0V, it changes once;

One double edge: one rising edge plus one falling edge;

High level: The voltage across TRIG+ and TRIG- keeps continuously 5~24V;

Low level: the voltage across TRIG+ and TRIG- keeps continuously 0V;

3 Hardware trigger (external trigger)

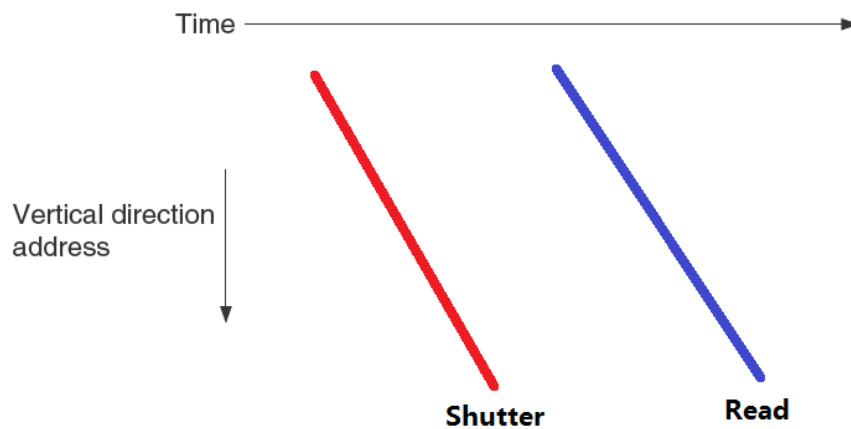
3.1 Trigger mode

- Edge trigger mode. You can select the upper edge or the lower edge as the trigger condition on the interface. When a valid trigger signal is received on the trigger terminal, the camera starts to collect a frame of image and transmit it to the host. Each valid edge corresponds to a trigger, and only one frame of image is output. Before the last frame of image acquisition is finished, the repeated trigger signal will be ignored. For example, if the camera's exposure time is set to 10 milliseconds, and the effective trigger signal is set to rising edge trigger, then within 10 milliseconds, even if there are multiple rising edge signals, only one frame of image will be triggered.
- Level trigger mode. You can select high level or low level as the trigger condition on the interface. When a trigger signal is at a valid level, the camera starts to collect a frame of image and transmits it to the host. After the frame is collected, if the signal is still at a valid level, the camera starts to collect the next frame of image and transmit it to the host, Until the trigger signal becomes inactive level.

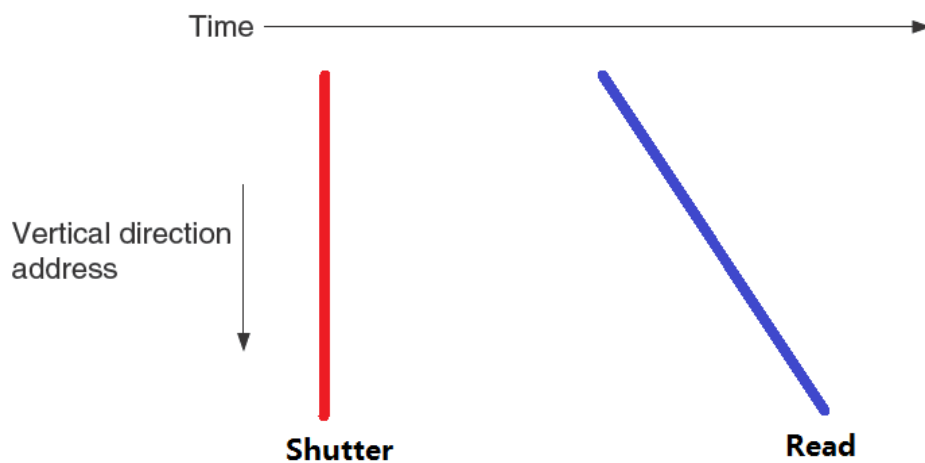
3.2 Shutter mode

There are three electronic shutter modes, namely ERS (rolling shutter), GRR (global reset), and GS (global exposure).

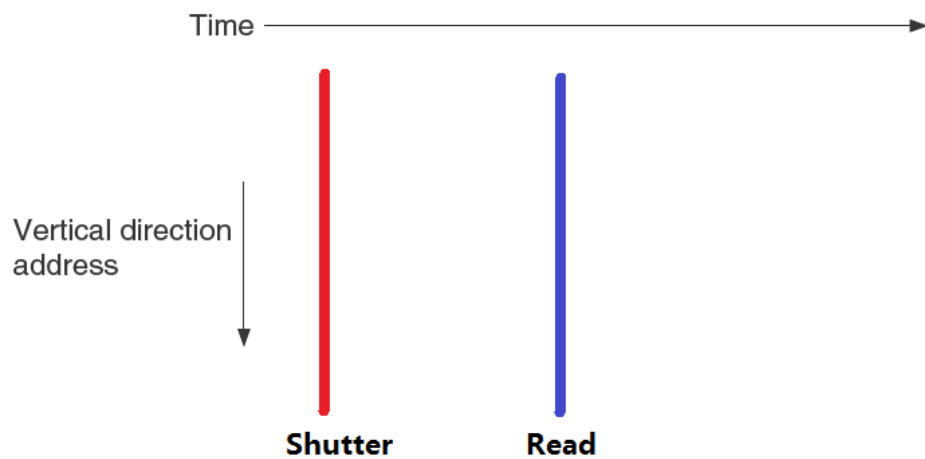
ERS: Rolling shutter exposure.



GRR: global reset, exposure is started at the same time, a rolling readout.



GS: Global exposure, simultaneous exposure and simultaneous reading.

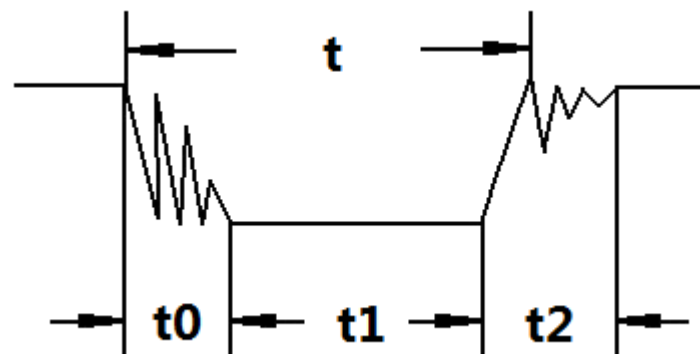


3.3 Trigger delay

Trigger delay means that when the external trigger signal is generated, the camera does not start exposure immediately, but starts exposure after a certain delay. When the delay is set to 0, it means no delay.

3.4 Eliminate trigger signal jitter

When a mechanical switch is connected to the external trigger terminal TRIG_IN, due to the jitter of the contact, it may cause false triggering. There is a hardware filter circuit in the camera, and the debounce time can be set to filter out jitter, such as t_0 and t_2 jitter in the following figure:



When the electronic switch is connected, debounce is not required, and the debounce time can be set to 0.

4 Flash sync mode

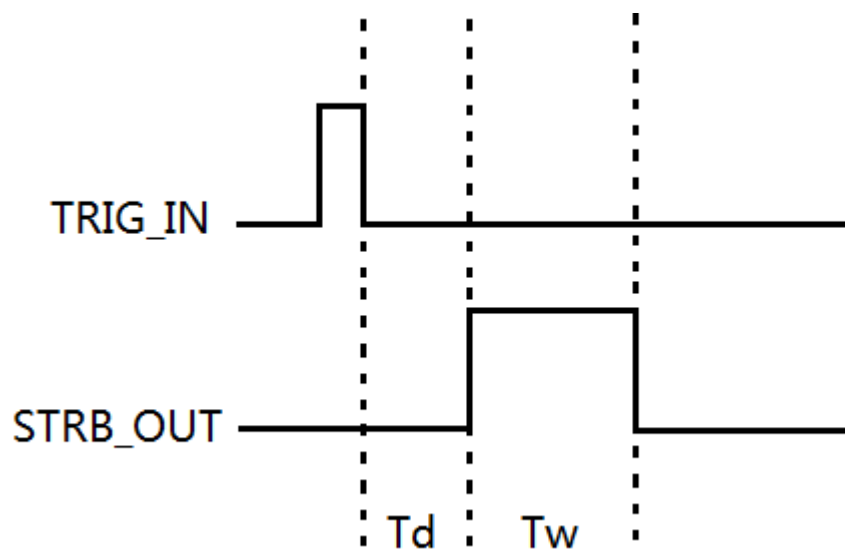
4.1 automatic

In the auto mode, the flash is synchronized with the exposure. When the sensor starts to expose, the flash turns on,

and when the sensor ends the exposure, the flash turns off. This mode is used by default.

4.2 Semi-automatic

In semi-automatic mode, the timing of the flash is completely controlled by software programming. It needs to set the delay and width. When an external signal is generated, the flash signal starts to output for a period. The timing is as follows (assuming that the trigger and flash output are both high and effective):



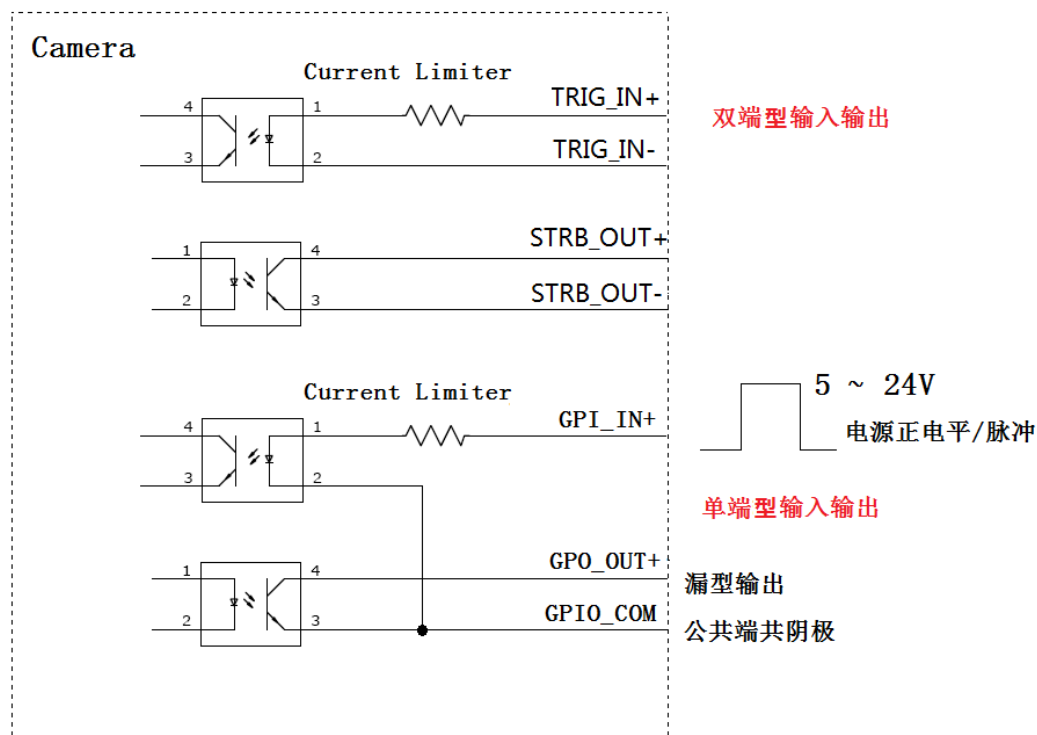
T_d is the delay, and T_w is the pulse width.

5 Trigger and GPIO principle and wiring

5.1 Schematic diagram

In the hardware, the input and output are all isolated by optocoupler, the input voltage is 5~24V; the output is open-drain, and the maximum current must not exceed 50mA.

The schematic diagram is as follows:

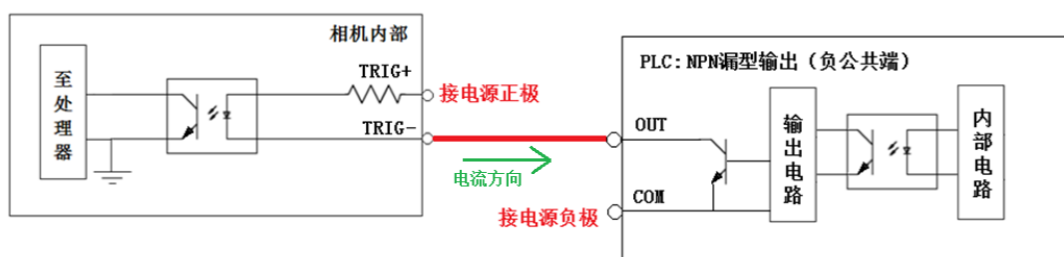


Note: Double-ended input can be connected to NPN and PNP switches, and double-ended output can drive common anode and common cathode loads; The single-ended input can only be connected to a PNP switch, the single-ended output can only drive a common anode load, and the single-ended common terminal can only be connected to the negative pole of the power supply.

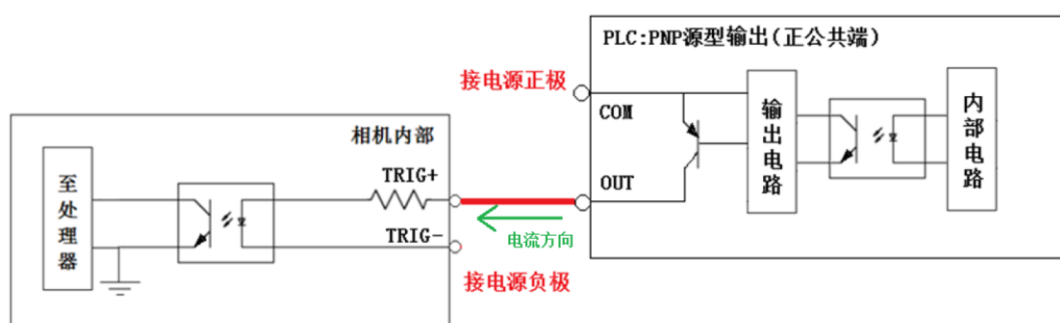
5.2 Wiring reference diagram

Note: The power supply is an external power supply. The voltage between the positive pole and the negative pole of the power supply is 5~24V. The red identification line is the wire; please refer to (6. Wire sequence definition) for the line color corresponding to the camera network name.

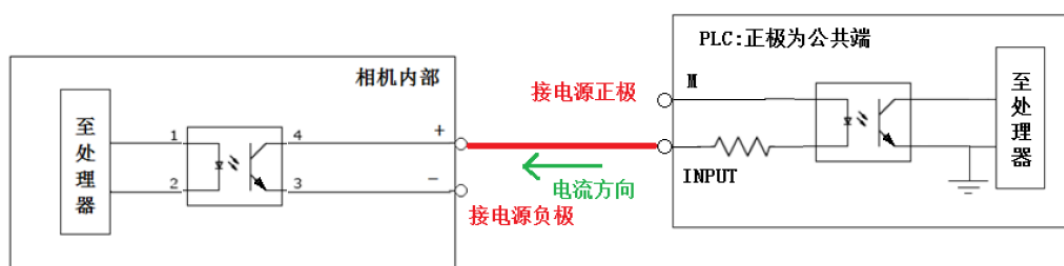
- (1) The camera input is connected to the NPN PLC output



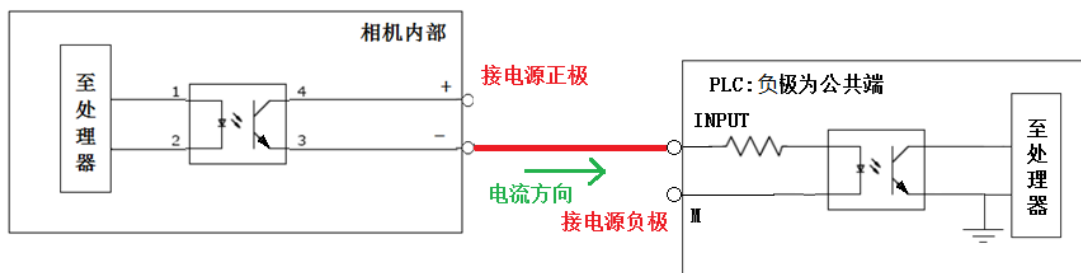
- (2) The camera input is connected to the PNP PLC output



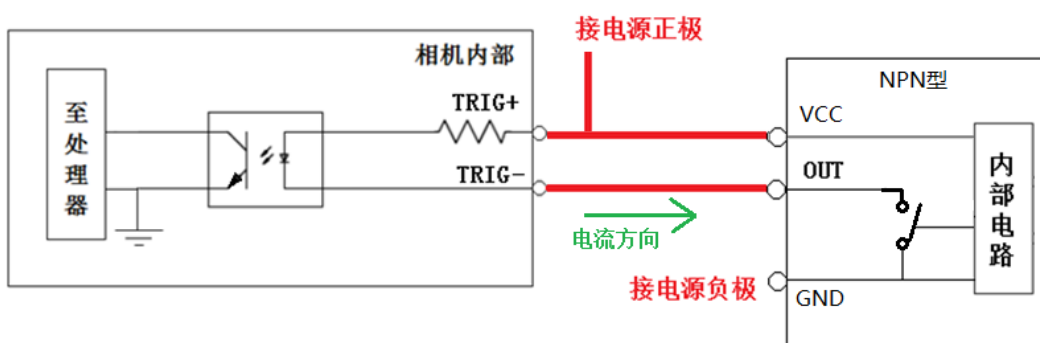
- (3) The camera output terminal is connected to the common anode PLC input terminal



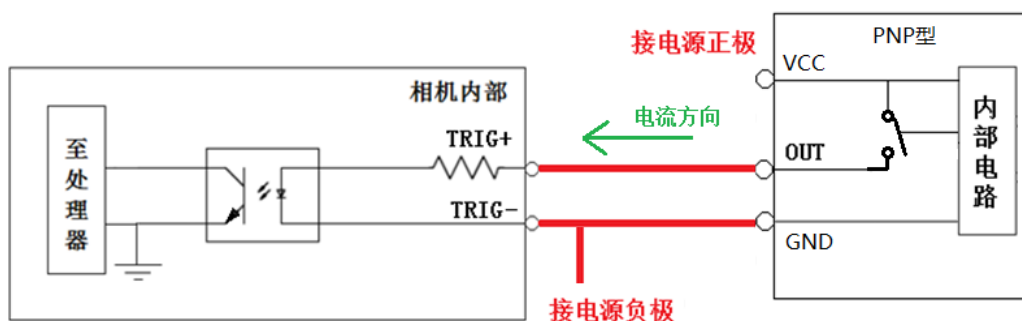
- (4) The camera output is connected to the common cathode PLC input



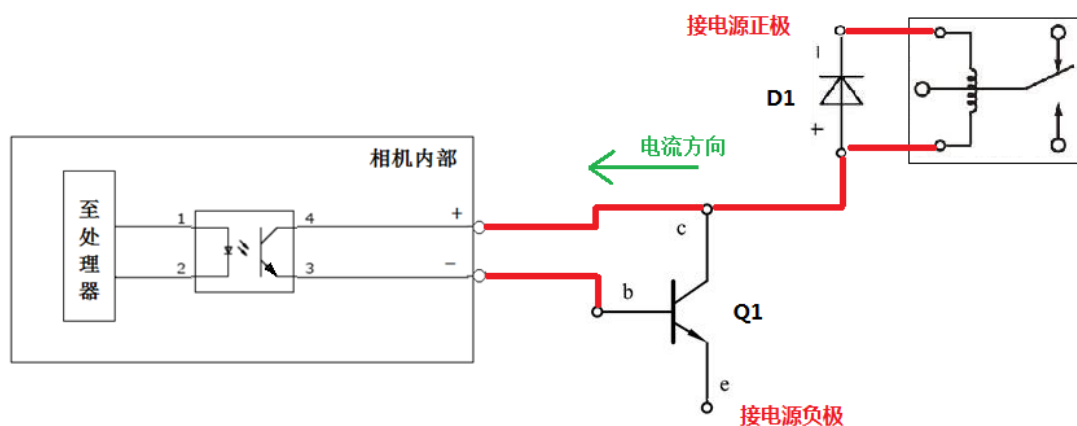
(5) The camera input is connected to the NPN photoelectric switch



(6) The camera input is connected to the PNP photoelectric switch



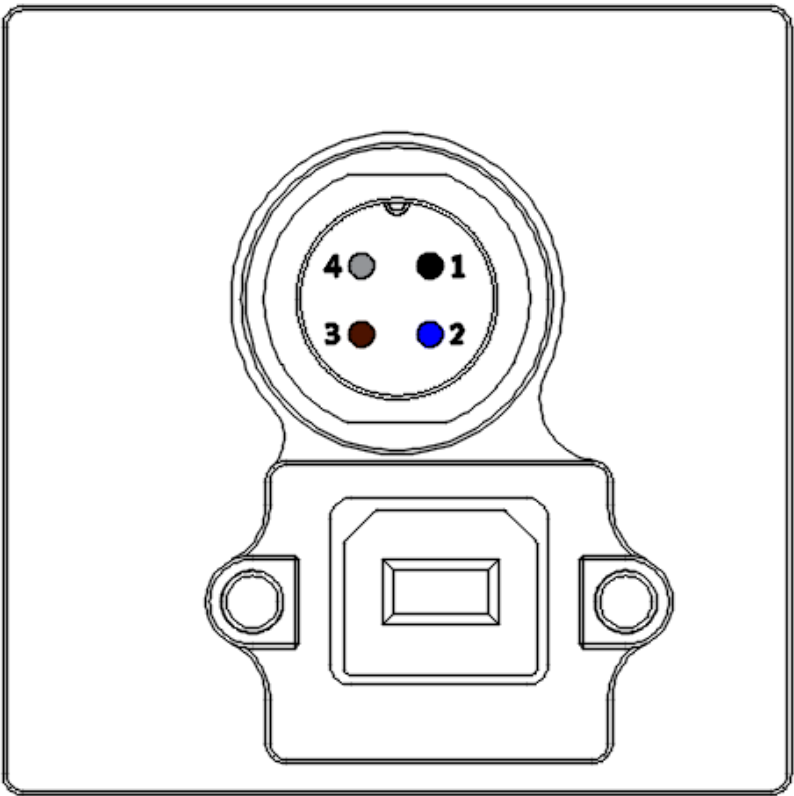
(7) The camera output terminal is connected to the relay (Q1 and D1 are external, reference model: Q1(C8050)D1(1N4148))



6 External trigger line sequence

definition

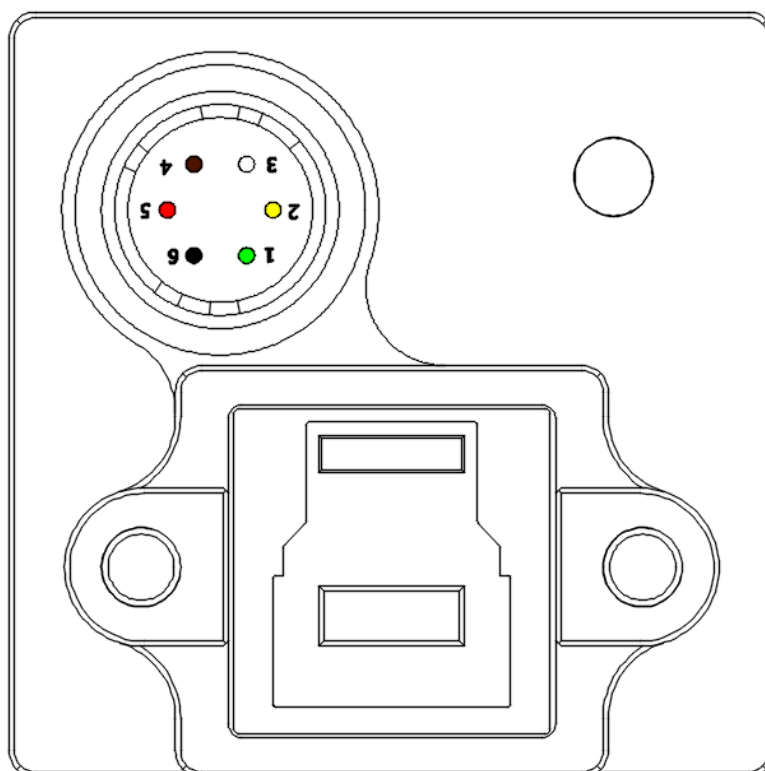
6.1 Four-core aviation head line sequence definition (UB series)



Pin number	Line color	Signal name	Description
1	black	TRIG_IN-	External trigger signal input negative terminal
2	blue	TRIG_IN+	External trigger signal input positive terminal

3	Brown	STRB_OUT-	Flash output negative terminal
4	gray	STRB_OUT+	Flash output positive terminal

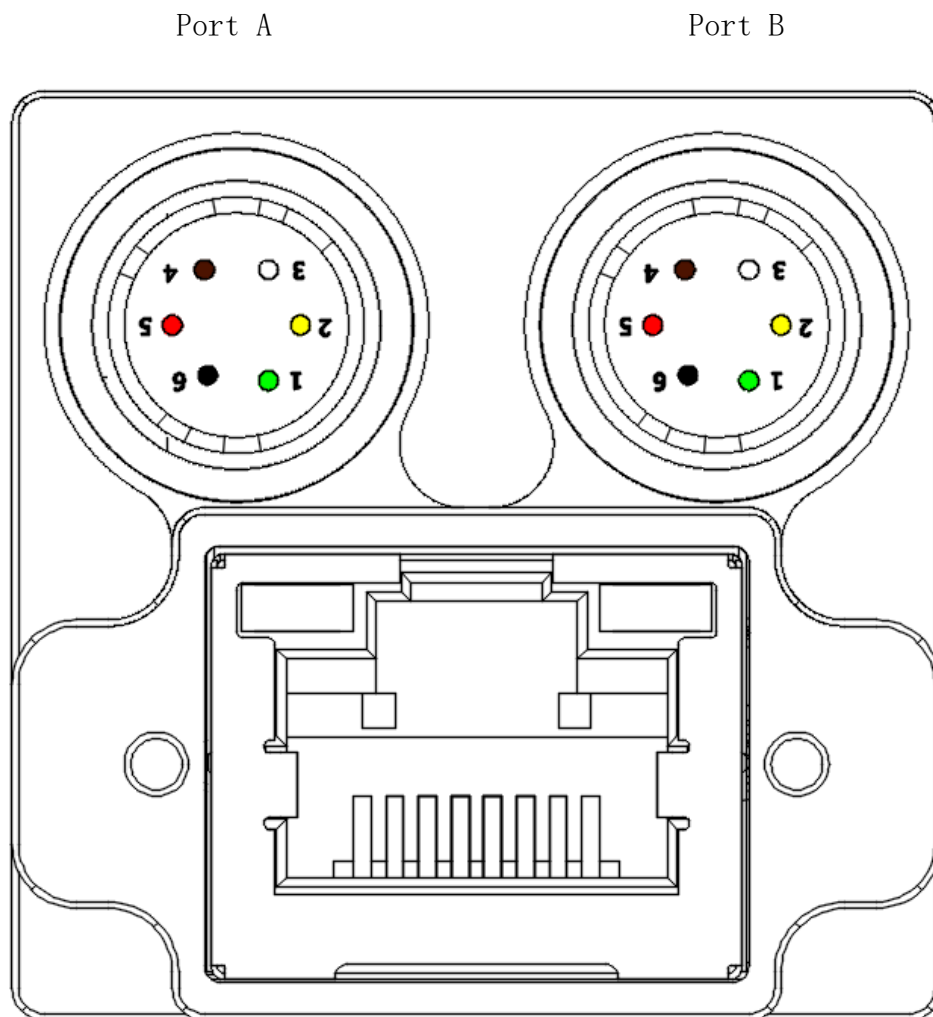
6.2 Six-core aviation head line sequence definition (UBS, SUA series)



Pin number	Line color	Signal name	Signal description	Remarks
1	green	GPO1+/STRB_OUT+	GPO1 positive terminal/flash output positive terminal	The default is flash output

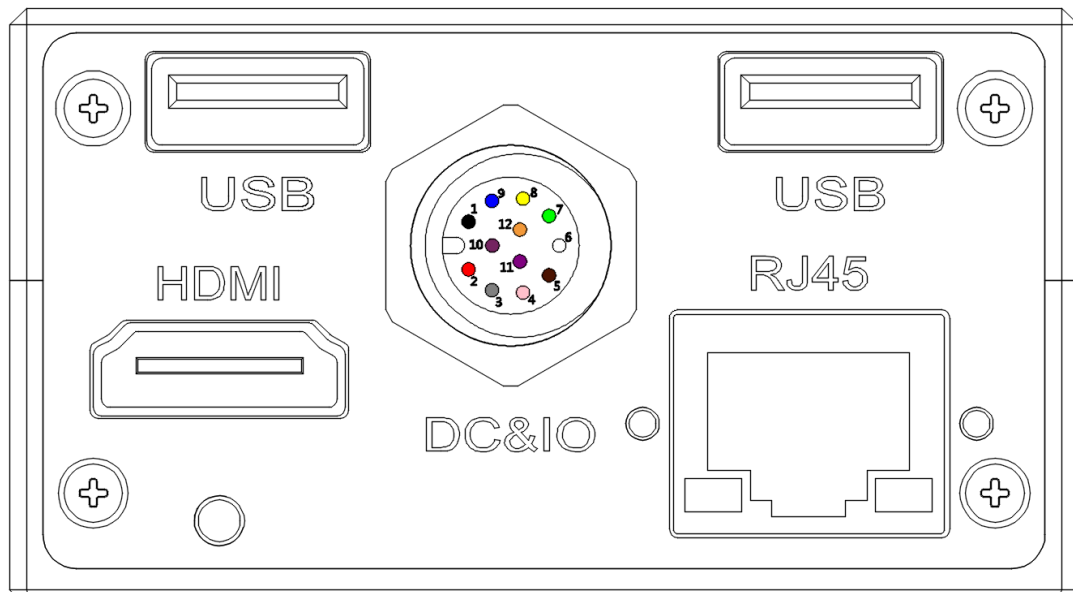
2	yellow	GPO1- /STRB_OUT-	GPO1 negative terminal/flash output negative terminal	The default is flash output
3	White	GPI1+ /TRIG_IN+	GPI1 positive terminal/trigger input positive terminal	Default is trigger input
4	Brown	GPI1- /TRIG_IN-	GPI1 negative terminal/trigger input negative terminal	Default is trigger input
5	red	GPO2+	GPO2 positive output	
6	black	GPO2-	GPO2 negative output	

6.3 Six-core aviation head line sequence definition (GE series)



port	Pin number	Line color	Signal name	Signal description	Remarks
Port A	1	green	GPO1+/STRB_OUT+	GPO1 positive terminal/flash output positive terminal	The default is flash output
	2	yellow	GPO1-/STRB_OUT-	GPO1 negative terminal/flash output negative terminal	The default is flash output
	3	White	GPI1+/TRIG_IN+	GPI1 positive terminal/trigger input positive terminal	Default is trigger input
	4	Brown	GPI1-/TRIG_IN-	GPI1 negative terminal/trigger input negative terminal	Default is trigger input
	5	red	PWR12V	Camera power input positive terminal	
	6	black	PWRGND	Camera power input negative terminal	
Port B	1	green	GPO2+	GPO2 positive output	
	2	yellow	GPO3+	GPO3 positive output	
	3	White	GPO4+	GPO4 positive output	
	4	Brown	GPI2+	GPI2 positive input	
	5	red	GPI3+	GPI3 positive input	
	6	black	GPIO_COM	GPIO public negative terminal	

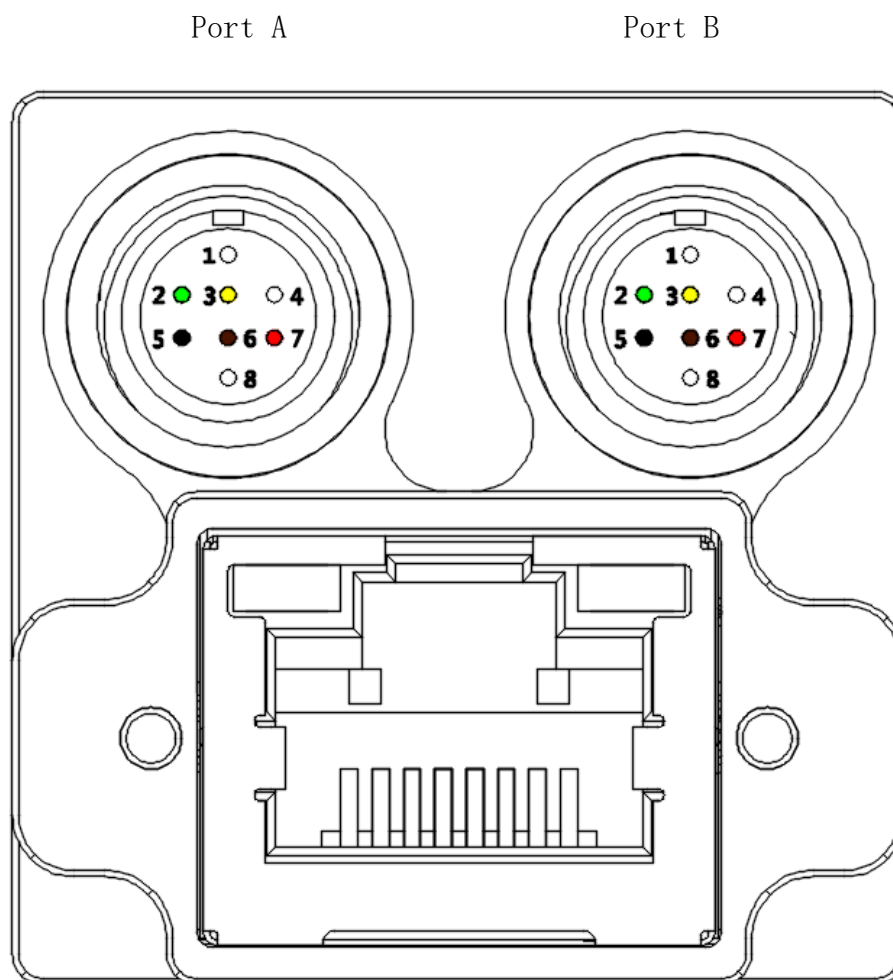
6.4 Twelve-core aviation head line sequence definition (ITA series)



Pin number	Line color	Signal name	Description
1	black	PWRGND	Camera power input negative terminal
2	red	PWR12V	Camera power input positive terminal
3	gray	UART_TX	Serial port send
4	pink	UART_RX	Serial port receiving
5	Brown	TRIG-	Trigger input negative terminal
6	White	TRIG+	Trigger input positive terminal
7	green	STRB+	Flash output positive terminal
8	yellow	STRB-	Flash output negative terminal

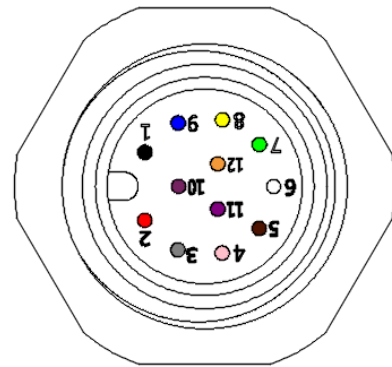
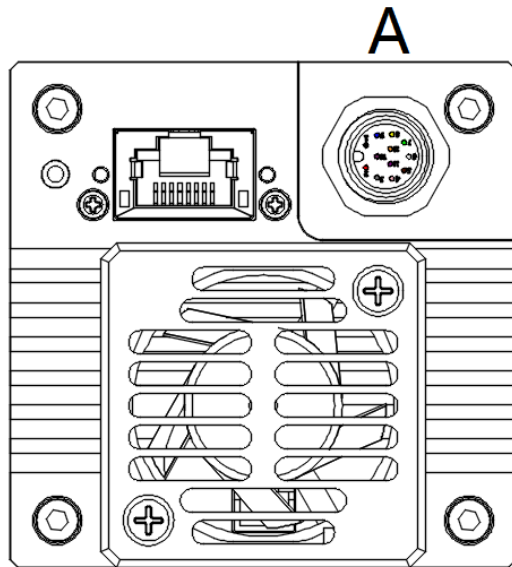
9	blue	GPIO_COM	GPIO public negative terminal/serial port ground
10	Magenta	GPO2+	GPO2 positive output
11	purple	GPO3+	GPO3 positive output
12	Orange	GPI2+	GPI2 positive input

6.5 Eight-core thread aviation head line sequence definition



port	Pin number	Line color	Signal name	Signal description	Remarks
Port A	1	White	GPI1+/TRIG_IN+	GPI1 positive terminal/trigger input positive terminal	Default is trigger input
	2	green	GPO1+/STRB_OUT+	GPO1 positive terminal/flash output positive terminal	The default is flash output
	3	yellow	GPO1-/STRB_OUT-	GPO1 negative terminal/flash output negative terminal	The default is flash output
	4	Empty feet			
	5	black	PWRGND	Camera power input negative terminal	
	6	Brown (High soft blue)	GPI1-/TRIG_IN-	GPI1 negative terminal/trigger input negative terminal	Default is trigger input
	7	red	PWR12V	Camera power input positive terminal	
	8	Empty feet			
Port B	1	White	GPO4+	GPO4 positive output	
	2	green	GPO2+	GPO2 positive output	
	3	yellow	GPO3+	GPO3 positive output	
	4	Empty feet			
	5	black	GPIO_COM	GPIO public negative terminal	
	6	Brown	GPI2+	GPI2 positive input	
	7	red	GPI3+	GPI3 positive input	
	8	Empty feet			

6.6 Twelve-core aviation head line sequence definition (XG series)

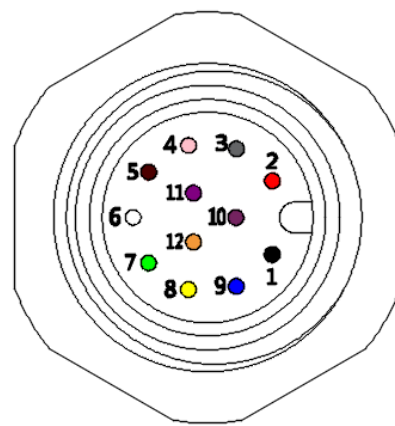
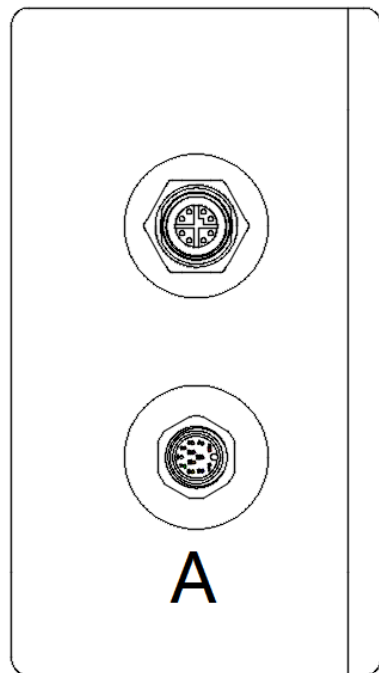


DETAIL A
SCALE 3:1

Pin number	Line color	Signal name	Signal description	Remarks
1	black	PWRGND	Camera power input negative terminal	
2	red	PWR12V	Camera power input positive terminal	
3	gray	GPI1+ / TRIG_IN+	GPI1 or trigger input optocoupler positive terminal	Default trigger
4	pink	GPI1- / TRIG_IN-	GPI1 or trigger input optocoupler negative terminal	Default trigger
5	Brown	GPI2+	GPI2 input optocoupler positive terminal	
6	White	GPI2-	GPI2 input optocoupler negative terminal	
7	green	GPI3+ / GPO3+	GPI3 input or GPO3 output optocoupler positive terminal	Default output
8	yellow	GPI3- / GPO3-	GPI3 input or GPO3 output optocoupler negative terminal	Default output

9	blue	GPO1+/STRB_OUT+	GPO1 or flash output photocoupler positive terminal	Default flash
10	Magenta	GPO1-/STRB_OUT-	GPO1 or flash output optocoupler negative terminal	Default flash
11	purple	GPO2+	GPO2 output optocoupler positive terminal	
12	Orange	GPO2-	GPO2 output optocoupler negative terminal	

6.7 Twelve-core aviation head line sequence definition (I3D series)



DETAIL A
SCALE 5:1

Pin number	Line color	Signal name	Description
1	black	PWRGND	Camera power input negative terminal
2	red	PWR12V	Camera power input positive terminal
3	gray	UART_TX	Serial port send
4	pink	UART_RX	Serial port receiving

5	Brown	TRIG-	Trigger input negative terminal
6	White	TRIG+	Trigger input positive terminal
7	green	STRB+	Flash output positive terminal
8	yellow	STRB-	Flash output negative terminal
9	blue	GPIO_COM	GPIO public negative terminal/serial port ground
10	Magenta	GPO2+	GPO2 positive output
11	purple	PWR12V	Camera power input positive terminal
12	Orange	PWRGND	Camera power input negative terminal